EQUATIONS OF LINES IN TWO SPACE:

In order to determine a straight line it is enough to specify either:

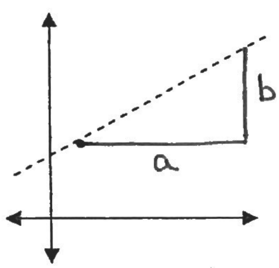
1. 2 points on the line

OR

2. 1 point on the line and its direction.

For a line, l, a fixed vector, $\vec{d}$, is called a **direction vector** for the line if it is parallel to l.

Note: every line has an infinite number of direction vectors that can be represented as $t\vec{d}$ where $\vec{d}$ is one direction vector for the line and t is a non-zero real number.



Any vector that is parallel to a line may be used as a direction vector for the line.

A line with direction vector $[d_1, d_2]$ has slope $\frac{b}{a}$, $a \neq 0$

Example: Find a direction vector for each line:

a) The line l_1 through points $A(4,-5)$ and $B(3,-7)$.

$\vec{d} = \vec{AB}$

$= [-1, -2]$

note: Any scalar multiple of $[-1, -2]$ could be a direction vector for l_1.

b) The line l_2 with slope $\frac{4}{5}$. $= \frac{b}{a}$

$\vec{d} = [a, b]$

$\vec{d} = [5, 4]$

note: A line with slope $= \frac{4}{5}$ that passes through the origin would pass through $(5, 4)$

$\therefore \vec{d} = [5-0, 4-0]$

$\vec{d} = [5, 4]$

The **Vector Equation of the Line** through the point of the line with position vector, $\vec{r}_1$, in the direction of $\vec{d}$ is of the form:

$$\vec{r} = \vec{r}_1 + t \cdot \vec{d} \quad , t \in \mathbb{R}$$

Example: Find vector equations of the line through points $A(1,7)$ and $B(4,0)$.

$\vec{d} = \vec{AB}$

$\vec{d} = [3,-7]$

$\therefore \vec{r} = \vec{r}_1 + t\vec{d} \quad , t \in \mathbb{R}$

$\vec{r} = [1,7] + t[3,-7] \, , t \in \mathbb{R}.$

OR

$\vec{r} = [4,0] + t[3,-7] \, , t \in \mathbb{R}.$

The real number, t, is called a **parameter**, each number t specifies one point on the line and to each point in the line there corresponds one value of t.

Example: Find three points on the line with vector equation: $\vec{p} = [0,4] + t[5,-2]$

① Let $t = 0$

$(0,4)$

Let $t = 1$

$(0,4) + (5,-2)$

$= (5,2)$

Let $t = 2$

$(0,4) + (10,-4)$

$= (10,0)$

The **Parametric Equations of the Line** passing through (x_1, y_1) in direction $[d_1, d_2]$ are of the form:

$$x = x_1 + t \cdot d_1$$
$$y = y_1 + t \cdot d_2 \quad , t \in \mathbb{R}$$

Example: Find parametric equations of the straight line through the point $P_1(1,2)$ and having a direction vector $\vec{d} = [3,1]$.

$x = x_1 + t d_1 \quad ; \quad y = y_1 + t d_2$

$\boxed{x = 1 + 3t} \qquad \boxed{y = 2 + t} \, , t \in \mathbb{R}$

A **Symmetric Equation of the Line** through (x_1, y_1) with direction vector $[d_1, d_2]$ is

$$t = \frac{x - x_1}{d_1} = \frac{y - y_1}{d_2}$$

Example: Find a symmetric equation of the line through points $A(2,-3)$ and $B(1,0)$.

$$\vec{d} = \vec{AB} = [-1, 3] \Rightarrow d_1 = -1 \; ; \; d_2 = 3$$

$$\therefore \quad t_1 = \frac{x-2}{-1} = \frac{y+3}{3}$$

OR

$$t_2 = \frac{x-1}{-1} = \frac{y}{3}$$

Example: Are the lines by the following vector equations coincident? That is, do these equations represent the same straight line?
$\vec{r} = [3,4] + s[2,-1]$ and $\vec{r} = [-9,10] + t[-6,3]$

* $\vec{d_1} = [2,-1]$; $\vec{d_2} = [-6,3]$

$\Rightarrow \vec{d_2} = -3\vec{d_1}$ (scalar multiple $\therefore l_2 \parallel l_1$)

* l_1 & l_2 are coincident if a point on one line satisfies the vector equation of the other line.

* pt $(3,4)$ is on l_1.

* If $(3,4)$ is on l_2, then $(3,4) = (-9,10) + t(-6,3)$ must be satisfied.

$$\left.\begin{aligned} 3 &= -9 - 6t \\ t &= -2 \end{aligned}\right\} \quad \begin{aligned} 4 &= 10 + 3t \\ t &= -2 \end{aligned}$$

$\therefore$ since t values are equal the lines are coincident.

EQUATIONS OF LINES IN THREE SPACE:

Vector Equations of the Line through the point with position vector $\vec{r}_1$, in direction $\vec{d}$ is of the form:

$$\vec{r} = \vec{r}_1 + t\vec{d}, \quad t \in \mathbb{R}.$$

Parametric Equations of the Line through the point (x_1, y_1, z_1) in the direction $[d_1, d_2, d_3]$ is of the form:

$$x = x_1 + td_1, \quad y = y_1 + td_2, \quad z = z_1 + td_3, \quad t \in \mathbb{R}.$$

Symmetric Equations of the Line through (x_1, y_1, z_1) with direction $[d_1, d_2, d_3]$ is of the form:

$$\frac{x - x_1}{d_1} = \frac{y - y_1}{d_2} = \frac{z - z_1}{d_3}$$

Example: Find the vector, parametric and symmetric equations of the line through the points $P_1(3,2,4)$ and $P_2(-1,7,2)$. Does the $Q(-3,5,2)$ lie on this line?

$$\vec{d} = \overrightarrow{P_1P_2} = [-4, 5, -2]$$

V.E. Line: $\vec{r} = [3,2,4] + t[-4,5,-2], \quad t \in \mathbb{R}.$

P.E. Line: $x = 3 - 4t, \quad y = 2 + 5t, \quad z = 4 - 2t, \quad t \in \mathbb{R}$

S.E. Line: $\dfrac{x-3}{-4} = \dfrac{y-2}{5} = \dfrac{z-4}{-2}$

$$Q(-3,5,2) \Rightarrow (-3,5,2) = (3,2,4) + t(-4,5,-2)$$

$-3 = 3 - 4t$ $5 = 2 + 5t$ $2 = 4 - 2t$

$t = \frac{3}{2}$ $t = \frac{3}{5}$ $t = 1$

* Since t-values are not all equal Q is not on the line.